Predicting and Removing Energy Gain in One-Sweep Contact Coupling

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Abstract

Position-based engines resolve contacts under a small, fixed local iteration budget, and a cheap way to make a stiff prop ring is to carry a few global modal amplitudes as extra unknowns in the unilateral contact row. At a few iterations per substep such a row can add energy rather than remove it. That it happens is known and not our claim; practitioner accounts report it as a tuning hazard without a framework. We supply that framework for the shared contact-to-modal row, parameterized by the host's velocity reconstruction $\dot{q}^+ = \kappa\,\Delta q/h$, which one cold read-back of $\Delta q/h$ against $2\Delta q/h$ identifies. A pre-solve danger index ρ, a ratio of two quadratic forms a host already assembles, predicts whether one mass-only sweep injects; the sign boundary is $\kappa^2 + (\omega h)^2 = 2 + m/M$. Charging the restorative block at the reconstruction-matched operator $\kappa^2 M_c + h^2 K_c$, in both the denominator and the correction, makes one sweep passive for any $\kappa \neq 0$ (rigid read-back at $\Delta z/h$) and any number of coupled coordinates, and at $\kappa = 1$ with $\zeta = 0$ it reproduces the converged implicit step exactly. Reconstruction dependence is a main finding: the mass-only and backward-Euler-shaped weights inject at low stiffness on the shipped symplectic host ($\kappa = 2$); the matched one does not. The closed forms are machine-checked to 10^{-12} relative on a 58,081-cell phase map and a nonmodal mass-spring collapse, and in exact rational arithmetic for the operator identity; on a shipped contact row the matched weight is passive on all 27 cells, and a three-arm weight swap takes the modal-energy overrun count from 8 of 24 to 0. Every theorem is confined to one sweep from a cold start with $e = 0$ and one normal-only row.

CCS Concepts

• **Computing methodologies → Physical simulation**.

Keywords

contact, position-based dynamics, energy, passivity, modal reduction, fixed-budget solvers

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1 Introduction

A rigid-body position-based solver [18] already resolves contacts and joints under a small, fixed local-iteration budget. To make a stiff prop such as a shelf ring when struck, the light option is a precomputed modal basis of a few global amplitudes [13, 21] carried as extra unknowns in the contact row that already couples the bodies: one host steps everything for a handful of runtime numbers.

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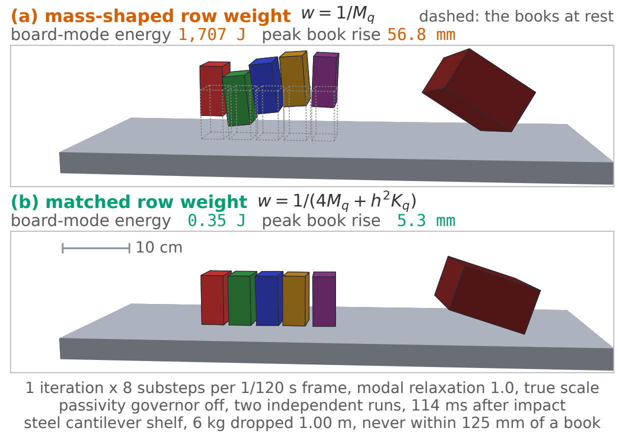

Figure 1: One shelf, one 6 kg weight dropped 1.00 m, one fixed budget (1 iteration × 8 substeps per 1/120 s frame); the arms differ only in the modal contact row's weight. (a) The mass-shaped weight puts the row's danger index above one and throws the five resting books off the shelf, peak rise 56.8 mm, at 1,707 J of board-mode energy, not book kinetic energy. (b) The matched charge $m_{\text{eff}} = m(\kappa^2 + (\omega h)^2)$ at this host's symplectic default $\kappa = 2$, priceable before the solve, holds them to 5.3 mm and 0.35 J. Fixed-budget injection is a known effect [9, 25]; the weight is what differs. A converged run on the same grid lifts a book 9.6 mm, so (b) is not ground truth.

The difficulty is energetic (Fig. 1). A hard-constraint position correction taken at a fixed iteration budget "can inject an arbitrary amount of energy" [10], and finite-iteration coupling can inject spurious energy even when each coupled subsystem is passive on its own [25]. Both are established observations, and this paper claims neither. What a practitioner lacks is a way to predict, for a given row, whether one sweep adds or removes energy, and what weight prevents it. A recent solver comparison records the symptom without a framework: position stabilization "can lead to energy creation and jitter if not tuned carefully" [6].

This paper is that framework for the shared unilateral contact-to-modal row between a rigid body of mass M and one restorative modal coordinate of effective mass m and frequency ω, on the stiffness-blind mass-normalized weight that position-based hosts default to. Every theorem is confined to one Gauss-Seidel sweep from a cold start, with restitution $e = 0$ and hard contact except where a compliance term appears. Within that regime we contribute:

- **C1.** A reconstruction-general one-sweep theorem (Thms. 3.1 and 3.3). For $\dot{q}^+ = \kappa\,\Delta q/h$ the mass-only sweep injects exactly when $\kappa^2 + (\omega h)^2 > 2 + m/M$; charging the restorative block at the matched operator $\kappa^2 M_c + h^2 K_c$, in both denominator and correction, makes one sweep passive for every

$\kappa \neq 0$ and any number of coupled coordinates, and under hard contact it loses exactly a perfectly inelastic impact. Per mode this is $m(1+(\omega h)^2)$ at $\kappa = 1$, where one undamped matched sweep is exactly the converged implicit step, and $m(4+(\omega h)^2)$ at the shipped symplectic default $\kappa = 2$.

- **C2.** The row-visible spectral form of that boundary at $\kappa = 1$, $h^2 JM^{-1}KM^{-1}J^\top \succ JM^{-1}J^\top$ (Thm. 3.2), reduced to a scalar danger index ρ a host can evaluate per row before it solves; $\rho > 1$ is injection.
- **C3.** A host-classification recipe and the relaxation boundary (Rmk. 1, Sec. 6): one cold sweep read back as $\Delta q/h$ or $2\Delta q/h$ fixes κ and hence both the correct index and the matched weight, while under-relaxing the modal correction by θ moves the boundary to $\theta^2(\kappa^2+(\omega h)^2) = 2 + m/M$.
- **C4.** An ordering result (Prop. 3.5): a trailing restorative row divides the one-sweep modal deposit by $(1+(\omega h)^2)^2$ at every κ, which makes that order passive at $\kappa = 1$ but at $\kappa = 2$ only narrows injection to a low-stiffness band.

The diagnostic and the weight are row-local, not a global safety mechanism: neither alone makes the solver passive.

2 One-sweep model

Bodies and row. A rigid body of mass M presents a row-visible mobility w_r (the scalar $1/M$ for a translational row; $w_r = 1/M + j_a^\top I^{-1} j_a$ for a full six-degree-of-freedom body). A restorative modal coordinate q has effective mass m, stiffness $k = m\omega^2$, and damping $2\zeta\omega m$. The substep is h; we write $b = (\omega h)^2 = h^2 k/m$ throughout. The unilateral hard contact row is $C = z - q \geq 0$, the rigid gap minus the surface deflection, with multiplier $\lambda \geq 0$.

Local solve and reconstruction. One Gauss-Seidel local solve of the compliant row is

$$\Delta\lambda = \frac{-C - \tilde{\alpha}\lambda}{w + \tilde{\alpha}}, \qquad \Delta x = M^{-1}J^\top \Delta\lambda, \tag{1}$$

with $\tilde{\alpha} = \alpha/h^2$ the compliance and w the row's weighted mobility. Position-based hosts recover velocity from the position change; backward Euler gives $\dot{q}^+ = \Delta x/h$. Below, κ rescales the *restorative* read-back only; the rigid block recovers $\Delta z/h$, as the shipped host does. One compliant row solved once from rest is the exact backward-Euler step for its own spring, relaxing a predicted $\tilde{q}$ to $\tilde{q}/(1+b)$ [15].

Cold start and ledger. A cold start has $q = \dot{q} = 0$, the contact exactly touching, incoming gap rate $v < 0$, and predicted penetration $\tilde{C} = hv$. Energies are the true Hamiltonian

$$E = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}m\dot{q}^2 + \tfrac{1}{2}kq^2 \tag{2}$$

at substep boundaries, and the sweep is *injecting* when $E^+ > E^-$. Since a passive unilateral contact cannot raise E and the converged step is dissipative [2], any one-sweep increase is a truncation artifact by definition, not a property of the contact model.

Scope (stated once, binds every theorem). One Gauss-Seidel sweep; cold start; restitution $e = 0$; hard contact except where $\tilde{\alpha}$ appears; one coupling row; normal-only, no friction.

3 Results

Theorem 3.1 (Scalar sign boundary, C1). *Under the scope of Section 2, mass-only weight $w_m = M^{-1} + m^{-1}$ and backward-Euler*

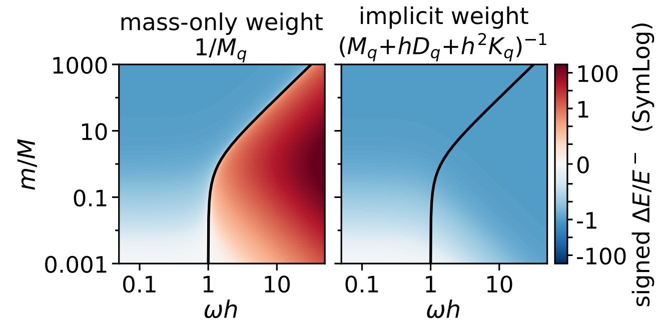

Figure 2: Phase map over $(\omega h,\ m/M)$. Left, the mass-only weight injects (red) exactly where the solid analytic curve $(\omega h)^2 = 1 + m/M$ predicts; right, the implicit weight is passive (blue). Hard contact, one sweep, cold start, $e = 0$, $\kappa = 1$.

reconstruction, one projection of the cold-start row satisfies

$$E^+ = \tfrac{1}{2}v^2\left(\tfrac{M}{M+m}\right)^2\left[\,M + m + m(\omega h)^2\,\right], \qquad E^- = \tfrac{1}{2}Mv^2, \tag{3}$$

and hence $E^+ > E^-$ if and only if $(\omega h)^2 > 1 + m/M$.

Proof. The cold prediction is $\tilde{C} = hv$, so (1) with $\tilde{\alpha} = 0$ gives $\Delta\lambda = -hv/w_m$ and, the relative normal rate being zeroed, the inelastic-pair reconstruction $v^+ = \dot{q}^+ = vM/(M+m)$, $q^+ = h\dot{q}^+$. Substituting into (2) and clearing $Mm > 0$ reduces $E^+ > E^-$ to $(\omega h)^2 > 1 + m/M$. ☐

At equality the projection is exactly energy-neutral; for $m \ll M$ the boundary is $\omega h = 1$. The offset $1 + m/M$ distinguishes it from the penalty-contact Courant limit at $(\omega h)^2 \sim O(1)$.

Theorem 3.2 (Row-visible condition, C2). *Let the row span a rigid block (mobility w_r) and modal coordinates with $a_i = U_i^2/m_i$ and $b_i = (\omega_i h)^2$. Under the scope of Section 2 with mass-only weights and $\kappa = 1$,*

$$\Delta E = \frac{v^2}{w_m^2}\left(\tfrac{1}{2}w_E - w_m\right), \quad w_m = w_r + \textstyle\sum_i a_i, \quad w_E = w_m + \textstyle\sum_i a_i b_i, \tag{4}$$

so the sweep injects if and only if

$$\textstyle\sum_i a_i b_i > w_r + \sum_i a_i \iff h^2 JM^{-1}KM^{-1}J^\top \succ JM^{-1}J^\top. \tag{5}$$

Writing $L = \sum_i a_i b_i = h^2 JM^{-1}KM^{-1}J^\top$, the row danger index $\rho = L/w_m$ satisfies $y := 2\,\Delta E\, w_m/v^2 = \rho - 1$ exactly, so injection is the single condition $\rho > 1$. A compliance $\tilde{\alpha}$ shifts it to $\rho = L/(w_m + 2\tilde{\alpha})$.

The right-hand operator of (5) is the Delassus, or inverse operational-space inertia, that contact solvers already form as an interface effective mass [19, 20]; the left-hand one reflects the material stiffness twice through the same row. Neither is new; the contribution is reading their ordering as the sign of one-sweep energy transfer. Because ρ lives in physical coordinates, (5) is not an artifact of any modal basis, as Section 4 confirms. Here J is one row, so $\succ$ is the scalar $L > w_m$; Thm. 3.3 gives the operator form for many coordinates sharing the row, and several *simultaneously active* rows are outside our scope, measured in Section 6.

Theorem 3.3 (Reconstruction-matched effective mass, C1). *Under the scope of Section 2 but with any number of coupled restorative* *coordinates, let the block have mass $M_c \succ 0$ and stiffness $K_c \succeq 0$, let the row price it at a charge operator W used in both the denominator of (1) and the position correction, and let the restorative velocity be reconstructed as $\dot q^+ = \kappa\,\Delta q/h$ and the rigid one as $\Delta z/h$. Write $G = \kappa^2 M_c + h^2 K_c$ and $u = W J_c$. Then one sweep changes the energy by*

$$\Delta E = \frac{v^2}{2D^2}\left[\, u^\top G\, u - w_r - 2J_c^\top u - 2\tilde\alpha \,\right], \qquad D = w_r + J_c^\top u + \tilde\alpha, \quad (6)$$

so it injects if and only if $u^\top G u > w_r + 2J_c^\top u + 2\tilde\alpha$. The reconstruction-matched charge $W = G^{-1}$ makes $u^\top G u = J_c^\top G^{-1} J_c$ telescope, giving

$$\Delta E = -\frac{v^2}{2D^2}\left(w_r + J_c^\top G^{-1} J_c + 2\tilde\alpha\right) \;<\; 0 \quad (7)$$

unconditionally, for every $\kappa \neq 0$, every $M_c \succ 0$, every $K_c \succeq 0$ and every $\tilde\alpha \ge 0$. At $\tilde\alpha = 0$, (7) is $-\frac{1}{2}[M m_{\mathrm{row}}/(M + m_{\mathrm{row}})]v^2$ with $m_{\mathrm{row}} = 1/(J_c^\top G^{-1} J_c)$, the energy lost in the perfectly inelastic impact of the pair (M, m_{row}).

For one coordinate and $\tilde\alpha = 0$, $u^\top G u > w_r + 2J_c^\top u$ reads $Mm(\kappa^2 + b) > 2M\mu_c + \mu_c^2$ for a scalar charge μ_c, the matched charge is $m_{\mathrm{eff}} = m(\kappa^2 + b)$, and (7) is $-\frac{1}{2}[M m_{\mathrm{eff}}/(M + m_{\mathrm{eff}})]v^2$: $m(1 + b)$ under backward Euler ($\kappa = 1$), $m(4 + b)$ under the symplectic default ($\kappa = 2$). Thms. 3.1 and 3.2 are its $W = M_c^{-1}$, $\kappa = 1$ specializations.

One scalar weight per mode realizes G^{-1} exactly when G is diagonal, that is whenever K_c is, as in a modal basis. The hypothesis is not cosmetic: charging only the diagonal of G when K_c has off-diagonal entries breaks (7), injecting on 19 of 2000 randomized coupled-stiffness rows where full G^{-1} stays passive on all. Diagonal, G^{-1} is r reciprocals per substep and leaves the row solve at 0.99 to 1.02 times the mass-only default; coupled, it needs one Cholesky of G, refactored per substep in our reference, and $2r^2$ flops per row against $2r$, 6.8 to 14.5 times the diagonal accumulation for $r = 4$ to 64, ratios on a reference implementation, not production timings.

Corollary 3.4 (Passive family, C1). *Under the hypotheses of Theorem 3.3 with W symmetric, one sweep is passive for every row J_c if and only if $0 \preceq W \preceq 2G^{-1}$ in the Loewner order. On the ray $W = cG^{-1}$, with $a = J_c^\top G^{-1} J_c \ge 0$, (6) collapses to*

$$\Delta E = \frac{v^2}{2D^2}\left[\, c(c-2)\,a - w_r - 2\tilde\alpha \,\right], \quad D = w_r + c\,a + \tilde\alpha, \quad (8)$$

so every $c \in [0, 2]$ is passive unconditionally, and outside that interval the row injects once $a > (w_r + 2\tilde\alpha)/[c(c-2)]$.

The matched charge is the interior point $c = 1$; the far endpoint $c = 2$ is the scalar charge $G/2$, which in one coordinate at $\kappa = 2$ is $m(2 + b/2)$: the converged amplitude that $c = 1$ misses (Section 4) is recovered exactly and passively there. Of 2500 randomized symmetric charges met with adversarial rows, all 1222 inside the interval are passive and all 1278 outside inject (T14a).

(6) accounts for the position *projection* only and E is the undamped Hamiltonian: the damper acts in the predictor, so its non-positive contribution is neither charged nor relied on; the bound is the conservative half of the step. E counts the bodies only: a compliant row also stores $\tilde\alpha v^2/(2D^2)$ in its own spring, so counting that term tightens each $2\tilde\alpha$ threshold to $\tilde\alpha$, a band on which the displayed ledger is permissive; no sign reported here changes, and the matched charge stays passive either way, at $-(v^2/2D^2)(w_r + a + \tilde\alpha)$. The two masses damping produces must not be conflated: the matched charge $m_{\mathrm{eff}} = m(\kappa^2 + b)$ is the *stored-energy* mass, damping-independent, and keeps (7) exact at any ζ because a cold predictor is inert and the damper stores nothing, whereas the *per-impulse* mass, $m(1 + 2\zeta\omega h + b)$ at $\kappa = 1$ and $m(1 + \zeta\omega h + b/4)$ at $\kappa = 2$, is strictly inside passivity at $\kappa = 1$ and unmatched at $\kappa = 2$. The loss form of (7) is the classical reduced-mass shock loss [17], $(M + hD + h^2K)^{-1}$ is the linearized implicit-step mobility whose stiffness term solvers routinely drop [1, 24], and $1 + (\omega h)^2$ is the backward-Euler propagator; we claim none of these. Ours is the effective mass a truncated row charges under general reconstruction, and the identity that one matched sweep is exactly that inelastic impact. The converged step being already passive [2], the statement lives only in the truncated regime.

Under the shipped convention the relaxation θ multiplies only the modal position correction, leaving the multiplier and rigid update at full strength. The modal deposit then scales as θ^2 with the rigid loss unchanged, so the mass-arm boundary becomes $\theta^2(\kappa^2 + b) > 2 + m/M$: under-relaxation raises the threshold and can move a marginally injecting sweep to passive, never the reverse. Relaxing the whole correction, rigid and modal, gives $L > (2/\theta - 1)w_m$.

Proposition 3.5 (Ordering divisor, C4). *Take two rows on (z, q): the contact $C_c = z - q$ (hard, unilateral) and a compliant spring $C_s = q$ with $\tilde\alpha_s = 1/(kh^2)$. At $n = 1$ from a cold start, order A = [contact, spring] and order B = [spring, contact] deposit modal energies D_A, D_B with $D_B/D_A = (1+b)^2$ at every κ, and order A injects exactly when $(\kappa^2 + b)/(1 + b)^2 > 2 + m/M$.*

Order B leaves the spring a no-op from rest and reproduces Thm. 3.1; order A deposits a fresh q_c that the trailing spring relaxes to $q_c/(1 + b)$, cutting the deposit by $(1 + b)^2$, so its injection condition is Thm. 3.1's divided by that factor. At $\kappa = 1$ the left side is $1/(1+b) \le 1 < 2 + m/M$, so that order is one-sweep passive at every stiffness and mass ratio. The add-on does not survive $\kappa = 2$: with $s = 2 + m/M$ the condition $sb^2 + (2s-1)b + s - \kappa^2 < 0$ has a positive root b^* exactly when $m < (\kappa^2 - 2)M$; order A then injects on $b < b^*$ by at most $(\kappa^2 - 2)^2/[4(\kappa^2 - 1)]$ of the incoming kinetic energy. At $\kappa = 2$ that is $m < 2M$, with $b^* = (\sqrt{37} - 5)/6 = 0.180$ at $m = M$ and a third of E^- at worst. That Gauss-Seidel order affects energy is known [4, 7], and the compliant-row-equals-backward-Euler premise is Macklin et al.'s [15]; the closed-form $(1 + (\omega h)^2)^2$ divisor and the reconstruction-scoped condition are ours.

Remark 1 (Reconstruction dependence). Thms. 3.1 and 3.2 are stated at $\dot q^+ = \Delta q/h$. The shipped host default is a symplectic implicit-midpoint stepper reconstructing $\dot q^+ = 2\Delta q/h - \dot q^n$, that is $2\Delta q/h$ at a cold start (measured exact on the shipped arrays), depositing four times the backward-Euler modal kinetic energy. Which block carries κ decides the boundary: scaling the rigid read-back by κ_r too weights the w_r term of (6) by $\kappa_r(2 - \kappa_r)$ and the $J_c^\top u$ and $\tilde\alpha$ terms by κ_r, so mass-only injection is $\kappa^2 + b > 2\kappa_r + \kappa_r(2 - \kappa_r)\,m/M$: $b > 1 + m/M$ at $(\kappa_r, \kappa) = (1, 1)$; $b > m/M - 2$ at $(1, 2)$, injecting at *every* stiffness once $m < 2M$; and $b > 0$ at $(2, 2)$. The matched charge is passive for every $\kappa \neq 0$ at Thm. 3.3's $\kappa_r = 1$, and under $\kappa_r = \kappa$ at every mass ratio if and only if $\kappa \in [1/2, 2]$. A weight matched to the wrong reconstruction is not passive either: the backward-Euler weight $m(1 + b)$ on a $\kappa = 2$ host injects whenever $M(2 - b) > m(1 + b)^2$, the low-stiffness injection measured in Section 4.

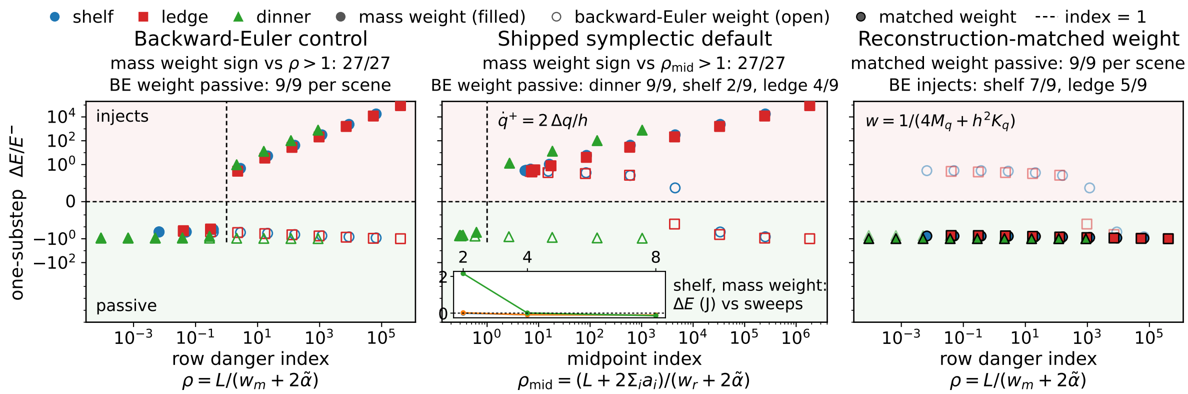

Figure 3: Shipped position-based contact row. Left, backward-Euler control. Middle, host default, where the reconstruction shifts the governing index to ρ_{mid} and the backward-Euler-shaped weight injects. Right, the matched weight $4M_q + h^2K_q$; inset, injection decays as the budget grows. Hard contact, one sweep, cold start, $e = 0$.

4 Validation

Relation to the companion study. A companion submission under review [3] is a system-level diagnostic: it measures injection and runs a weight-swap control on a production host, with no closed forms. This paper is the per-sweep closed-form theory. Shared evidence is confined to the host and to two arms of the 24-cell weight swap below, mass-only and $(M_q + hD_q + h^2K_q)^{-1}$; everything else is new, including the 27-cell shipped-row test and the matched arm.

Every closed form above was checked numerically by the suite of Table 1; the toy and nonmodal models are structurally independent code paths from the analytic forms, so a shared implementation error would not agree across them, and the operator identity is verified in exact rational arithmetic. The attached anonymous supplement ships every check in that table with its per-cell CSV: runnable code for the analytic arms, drivers and host excerpts for the shipped-host ones. The one loose tolerance is T4, the shipped row: its mass arm reproduces the backward-Euler identity to the 10^{-3} magnitude band, while the implicit arm is validated at sign and passivity level, its magnitude deviating up to 4.5×10^{-2} from two slack sources named in advance, quaternion second-order terms and the velocity pass.

Phase map (Fig. 2). A 241×241 grid in $(\omega h,\ m/M)$, measured by the one-sweep projection rather than the closed form, misclassifies no cell against $(\omega h)^2 = 1 + m/M$.

Nonmodal collapse (Fig. 4). We repeat the measurement in plain point-mass and spring coordinates, no modal basis anywhere, over four variants including a wide plank row that exercises the off-diagonal of K through $JM^{-1}KM^{-1}J^{\top}$. All 288 live cells agree in sign with $\rho > 1$, and every variant collapses onto $y = \rho - 1$ to 4.3×10^{-14}.

Ordering. The two-row toy confirms Prop. 3.5: at $n = 1$ the ratio $D_B/D_A = (1 + b)^2$ is exact, 10201 at $b = 100$. Carrying κ through the toy on 10,000 cells per reconstruction (T13) leaves that order passive on all at $\kappa = 1$ and injecting on 3840 at $\kappa = 2$, which at the shipped $h = 1/960$ s substep is every mode below 65 Hz at $m = M$. Both orders reach the same converged backward-Euler point, dissipative when read back at $\kappa = 1$, monotonically from opposite sides.

Accuracy against a converged reference. Passivity is a weak certificate: over-dissipating is passive too. We compare each arm against the exactly solved converged implicit contact step, on contact impulse, post-step rigid velocity, modal amplitude and modal velocity (T11). First, the constraint residual cannot discriminate: one sweep of a single row drives C^+ to zero for *every* weight, since $\Delta C = w\,\Delta\lambda = -C$, measured at 2.1×10^{-16} of $|hv|$ across 32,000 arm-cells. Second, at $\kappa = 1$ and $\zeta = 0$ the matched weight reproduces the converged backward-Euler step *exactly*, to 5.5×10^{-16} relative on all four quantities. With damping the amplitude gap is exactly $2\zeta\omega h/(1 + M/m + b)$, first order in $\zeta\omega h$ over four decades and $0.979\,\zeta\omega h$ at the measured $m = M$, $b = 0.043$. Third, still at $\kappa = 1$, the mass-only arm's defect

Table 1: Validation suite. *Model*: *analytic* a closed-form row, *nonmodal* a mass-spring chain, *shipped* a production solver. *Cells* counts independent parameter draws, each possibly run under several weights or reconstructions. *Crit.* is the per-cell test: a relative residual bound, *sign* for predicted vs. measured energy sign, *exact* for rational equality, or *ratio* for T6, where a cell *overruns* when its peak modal energy exceeds the impactor's peak rigid kinetic energy. Under *Result*, *checks* counts named assertions over a block's cells; other entries count cells, out of *Cells* unless a denominator is given; T4, T6, T13, T14b give one per arm. Signs are undefined where the predicted change vanishes; cells within 10^{-9} of it are non-live: none in T2, T5, T7, T13, 12 of T3's 300, all with $|\Delta E| \le 2.3 \times 10^{-16} E^-$. T4's pair uses a backward-Euler control; the host default is Fig. 3. Only T10's 300 and T14a's 720 rational cells are exact; their float cells are conditioning-limited at 10^{-9}.

	What	Model	Cells	Crit.	Result
T1	Identity battery	analytic	2626	10^{-12}	29/29 checks
T2	Phase map	analytic	58,081	10^{-12}, sign	0 misclass.
T3	Nonmodal collapse	nonmodal	300	10^{-12}, sign	288/288 live
T4	Shipped row	shipped	2×27	sign	27/27, 27/27
T5	Ordering, converged	analytic	208	10^{-12}	184/184 checks
T6	Weight swap	shipped	3×24	ratio	8/0/0 overrun
T7	Recon. mass law	analytic	7000	10^{-12}, sign	0 mismatch
T8a	Relax. boundary	analytic	1200	10^{-12}, sign	0 mismatch
T8b	Relax. direction	analytic	1600	sign	0 unsafe flips
T9	Matched arm	shipped	27	sign	27/27 passive
T10	Operator form	analytic	9900, 300	10^{-9}; exact	17/17 checks
T11	Converged reference	analytic	15,000	10^{-12}	8/8 checks
T12	Hypothesis pinning	analytic	53,000	10^{-12}, sign	28/28 checks
T13	Ordering vs. κ	analytic	2×10^4	10^{-12}, exact	0, 3840 inject
T14a	Charge family	analytic	2100, 720	10^{-9}; exact	0 inject in $[0, 2]$
T14b	Equal cost	shipped	12×27	sign	22 vs. 0 inject
T14c	Warm, two-row	analytic	43,783	sign	2795 unsafe

has an accuracy face: it over-deposits the modal amplitude by exactly $(M + m(1 + b))/(M + m) = 1 + b\,m/(M + m)$, which at $m = M$ is 51 at $b = 10^2$ and 5001 at $b = 10^4$, the same $(\omega h)^2$ that drives the sign boundary. Fourth, the reference at $\kappa = 2$ is the host's own converged step, the mode on implicit midpoint with the row at position level, so $v^+ = \dot{q}^+/2$ and the row reproduces it at $\mu_* = m(2 + b/2)$. The matched charge $m(4 + b)$ is exactly $2\mu_*$, so it over-dissipates: the amplitude ratio is $(M + \mu_*)/(M + 2\mu_*)$, falling to 1/2 only as $\mu_*/M \to \infty$, and it is never less dissipative than the reference on any of 3000 cells. Passive and conservative, not accurate.

Shipped contact row (Fig. 3). We drive one cold-start row of a production position-based support solver, the host of the companion study [3], on its three scenes, with the passivity governor off, relaxation at 1 and no other contacts, nine stiffness points per scene across many decades of ρ (27 cells), under both the host default reconstruction and a backward-Euler control. Under the control the mass-arm sign matches $\rho > 1$ on all 27 cells and the backward-Euler remedy is passive on all 27. Under the host default (Rmk. 1) the governing index is the $\kappa = 2$ instance of Thm. 3.3, which for mass-only weights reads $\rho_{\text{mid}} = (L + 2\sum_i a_i)/(w_r + 2\tilde{\alpha})$: $\sum_i a_i$ moves from the denominator to the numerator, so this is not a rescaling of ρ. It tracks the measured mass-arm sign on all 27 cells, whereas ρ tracks only 22, and the five misses are false *negatives*, the unsafe direction for a pre-solve check: a safe row that measurably injects, at the softest modes of the two strongly coupled scenes. The backward-Euler-shaped remedy, no longer matched to the reconstruction, stays passive only where the coupling is weak (dinner 9/9; shelf 2/9, ledge 4/9), while the matched weight $m(4 + (\omega h)^2) = 4M_q + h^2 K_q$ is passive on every cell, its danger index below one throughout, at the same operating points where the unmatched weight injects.

System-level corroboration. The 24-cell weight-swap grid of the companion study [3] runs a production host across three scenes, two relaxations $\theta \in \{0.7, 1.0\}$, and four iteration-substep budgets $(4, 1)$, $(8, 2)$, $(16, 4)$, $(32, 8)$, read as iterations $\times$ substeps per 1/120 s frame, so h runs from 1/120 to 1/960 s, with the symplectic stepper and the passivity governor off (hard contact; cold impacts onto warm, simultaneous support rows, beyond the one-sweep scope). We re-run it with a third arm. Under the stiffness-blind mass-only weight 8 of 24 cells overrun, their peak modal energy exceeding the measured rigid-side supply by up to 1.2×10^5 at the lowest budget. The ratio is a gross-overrun test; the total-energy sign gives 10, 1 and 0 of 24 across the three arms. Under the weight $(M_q + hD_q + h^2K_q)^{-1}$ none do (0 of 24), which is the companion's published arm; but that weight is shaped for $\kappa = 1$ while this host reconstructs at $\kappa = 2$, so by Thm. 3.3 it is *not* the matched weight here and its success is evidence about stiffening the row, not about the one-sweep identity. The arm that does test the theorem charges $4M_q + h^2K_q$, also overrun-free (0 of 24) and lying strictly below the $\kappa = 1$ arm on every cell, consistent with the over-dissipation predicted for $\kappa = 2$. The eight overrunning cells are exactly the two strongly coupled scenes at the two lowest budgets and both relaxations, matching Theorem 3.1's coupling-strength dependence. Our mass-only arm reproduces the companion's published ratios exactly on all 24 cells, so the two grids are the same experiment.

Is the weight worth its budget? At fixed stiffness injection decays to passivity by four to eight iterations (Fig. 3, inset), but across all 27 cells more slowly: iterating the mass-only weight leaves 17, 9, 6 and 2 cells injecting at 2, 4, 8 and 16. The last still overshoots one cell by 28 times its incoming kinetic energy at 5.3 times the measured substep cost, whereas the matched weight is passive on all 27 at 0.996 of the mass-only cost. What iterating converges to is the reference, not the guarantee: passive on all 27 at 64 iterations and 20 times the cost. The guarantee is paid for in amplitude instead, a median 0.57 of the converged against 0.85 at 16 iterations (T14b).

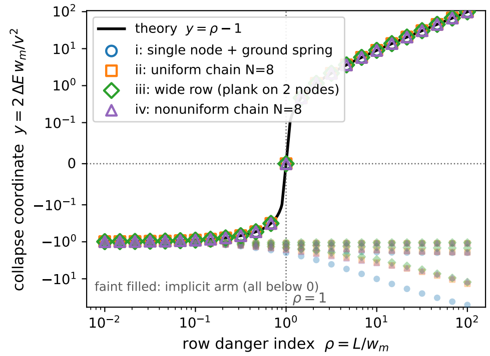

Figure 4: Nonmodal collapse: four mass-spring variants on $y = \rho - 1$ of Thm. 3.2; the faint implicit arm is negative.

5 Related work

Co-simulation stability and energy residuals. The two-mass oscillator is the canonical co-simulation stability model, but its boundaries are asymptotic spectral-radius conditions on a macro-step amplification matrix [11, 14], not per-sweep energy signs, and Arnold's sequential-coupling stability analysis [4] is a contractivity condition, not a closed-form deposit factor. Energy-sign framings of coupling error exist as runtime residuals for adaptive step control [22] and as energy-leak monitoring [12], again not parametric thresholds; the interface effective-mass operator behind our ρ serves elsewhere as a stabilizing predictor [20]. C1, C2 and C4 are the closed-form, per-sweep, unilateral-row content those programs lack.

Graphics constraint solvers. Position-based dynamics establishes the compliant-row premise and the stiffness-blind default weight [15], and single-iteration substepping targets energy loss, not gain [16]. Reduced deformable coordinates have been carried inside a position-based host as extra degrees of freedom [21], so stepping modal amplitudes in the solver is established practice; we claim neither the coupling nor the embedding. Augmented vertex block descent concedes hard-constraint injection and mitigates it empirically [9], building on Chen et al. [8], and energy-aware Gauss-Seidel studies within-solve energy [7], but none gives a closed-form sign boundary or a one-sweep passivity identity; practitioner accounts report the symptom and prescribe tuning [5, 6].

Contact mechanics endpoints and reduced coordinates. The reduced-mass shock loss [17] and the dissipativity of the converged solve [2] bound the endpoints our one-sweep statement sits between. Reduced-coordinate contact against internal modes has been posed as a momentum-conserving remapping without an energy bound [23], and energy targets set at the whole-system rather than per-row level [26]. The closest guarantee is finite-iteration coupling that builds in discrete passivity for any inner budget on bilateral interfaces [25]; ours is orthogonal, a sign diagnostic and weight for a unilateral row rather than automatic passivity.

6 Practitioner guidance and limitations

When to couple into the row. Coupling the mode into the row is one design point. A stiff prop can instead ring from an open-loop modal bank driven by the solver's own impulse, the standard route in contact-sound synthesis, which sidesteps injection: the mode never enters the row. In-solve coupling is needed when a second object rests persistently on the deforming region while it rings, cargo on a shelf or a plank on a ledge: the deflection is then at or above the contact tolerance, and a bank that never updates it prices support impulses against a stale surface. Open-loop excitation stays the safer default for audio-rate ringing and unloaded props.

Recipe. (i) Classify. Drive one cold row and read back the restorative velocity: $\Delta q/h$ is $\kappa = 1$, $2\Delta q/h$ is $\kappa = 2$; Thm. 3.3 reads the rigid block at $\Delta z/h$, and if the host rescales that block too, use Rmk. 1's boundary. *(ii) Test before solving:* with w_r, a_i, b_i and L of Thm. 3.2, mass-only weights inject exactly when $\rho_\kappa = [L + (\kappa^2 - 2)\sum_i a_i]/(w_r + 2\tilde{\alpha}) > 1$; ρ_{mid} is its $\kappa = 2$ instance, and evaluating it costs 6% of the row solve it precedes. *(iii) Charge* the restorative block at $W = (\kappa^2 M_c + h^2 K_c)^{-1}$ in the denominator of (1) *and* in the position correction; the denominator alone is not the proved method. *(iv) Factor once:* a diagonal K_c, as in a modal basis, makes $W\,r$ reciprocals with no matrix assembled; when K_c couples, one Cholesky of G is shared across rows and, the basis and h being fixed, hoists to load time. *(v) Choose the family point:* the matched charge is $c = 1$ of Cor. 3.4, per mode $m(\kappa^2 + (\omega h)^2)$. At a low budget it rings more weakly than the mass-only arm's over-deposit, so a stronger ring means iterating the row, decoupling to open-loop excitation, or moving toward $c = 2$, which returns the converged amplitude exactly where $c = 1$ returns 0.50 to 0.60 of it, at the cost of margin: headroom to the injection threshold falls under 1% on a fifth of the cells measured (median 36%) against at least 100% at $c = 1$ (median 171%). A mis-estimated G is the same family at $c\,G/\hat{G}$, so $c = 1$ absorbs any overestimate of G and a twofold underestimate, while $c = 2$ loses the guarantee under any underestimate; keep $c = 1$ unless the ring amplitude is the deliverable (T14a). Not matching leaves weak coupling ($2\sum_i a_i \le w_r$), iterating to convergence, or under-relaxing, a partial mitigation rather than a substitute.

Limitations. Section 2's scope binds every theorem, the reconstruction included: ordering the restorative row after the contact row is a passivity fix at $\kappa = 1$ only, and on the symplectic default it leaves the low-stiffness band of Prop. 3.5. Warm states and simultaneously active rows are measured, not proved (T14c): the mass-only weight itself injects on 19,730 of 43,783 warm and two-row cells, 2795 of them called safe by its index, and order alone flips 734 signs; the matched charge, clean under the shipped $\kappa = 2$ midpoint pairing and secularly bounded over 20,000 substeps of resting contact, still injects on 4099 of 43,898, a measured reduction and not a guarantee (the denominators differ because row activity after the sweep depends on the weight; both arms scan the same draws). The interior $1 < n < \infty$, restitution and friction are open.

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